Discrete-Time Quantum Mechanics: Unitarity, Lorentz Invariance, and Observable Predictions

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Submitted for review

April 21, 2026

Abstract

We develop a self-consistent framework for discrete-time quantum mechanics based on the Cayley (Crank–Nicolson) discretization of proper-time evolution at the Planck scale. After classifying all single-step unitary evolution operators as rational functions on the imaginary axis, we identify the $[1, 1]$ diagonal Padé approximant—the Cayley map—as the unique exactly unitary, spatially local, rational scheme compatible with Lorentz symmetry via a proper-time parameterization. We derive this scheme independently from the minimal two-point noncommutative spectral triple, establishing a structural, not merely computational, justification. The framework yields a calculable modified dispersion relation $\omega_{\mathrm{Cay}}(E) = (2/t_P)\arctan(t_P E/2\hbar)$, a universal subluminal group-velocity correction $\delta v/c = -E^2/(4E_P^2)$ whose sign distinguishes it from κ-deformed competitors, and a generalized uncertainty principle with correct limiting behaviour. We assess observational prospects honestly: direct GRB time-delay tests are out of reach by $\sim$17 orders of magnitude, but the sign and form of Planck-scale corrections constitute a falsifiable prediction. Connections to loop quantum gravity, causal set theory, and the AdS/CFT correspondence are discussed, and a programme of open problems is outlined.

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1 Introduction

The quantisation of spacetime at the Planck scale is among the deepest unresolved problems in theoretical physics. A recurring suggestion is that time itself may be discrete at the Planck length $\ell_P \approx 1.62 \times 10^{-35}$ m, with the Planck time $t_P \approx 5.39 \times 10^{-44}$ s playing the role of a fundamental tick [**?**, **?**, **?**]. Yet nearly every attempt to implement discrete-time quantum mechanics faces an immediate obstacle: naive discretization of the Schrödinger equation destroys the unitarity of time evolution, thereby violating probability conservation and undermining the probabilistic interpretation of quantum mechanics.

This paper addresses three intertwined questions:

1. **Unitarity.** Among all rational single-step discretization schemes, which are exactly unitary, and what is the unique simplest such scheme?

2. **Lorentz invariance.** If a fundamental time step $a = t_P$ exists, does it break Lorentz symmetry, and if so can that breaking be avoided?

3. **Observable predictions.** What are the leading Planck-scale corrections to dispersion relations, uncertainty principles, and propagation speeds, and how do they compare with competing frameworks?

The central proposal. We argue that the *Cayley (Crank–Nicolson) discretization* of proper-time evolution,

$$\left(\mathbf{1} + \frac{it_P H_\tau}{2\hbar}\right) |\psi(\tau + t_P)\rangle = \left(\mathbf{1} - \frac{it_P H_\tau}{2\hbar}\right) |\psi(\tau)\rangle, \tag{1}$$

where τ is proper time and H_τ is the rest-frame energy operator, is the unique scheme that is simultaneously (i) exactly unitary, (ii) compatible with Lorentz symmetry via proper-time parameterisation, and (iii) derivable from the minimal two-point noncommutative spectral triple. Its Planck-scale corrections are calculable, carry a universal sign (subluminal), and are physically distinguishable from κ-deformed competitors.

Metric signature and units. Throughout this paper we use the metric signature $(+---)$. The bulk of the theoretical development is carried out in *Planck units* $\hbar = c = G = 1$, so that $t_P = \ell_P = 1$ and the Planck energy is $E_P = 1$. When making contact with SI numbers for observational estimates we restore factors of $\hbar$, c, and G explicitly.

Organisation. Section 2 establishes the Hilbert-space framework and the unitarity classification theorem. Section 3 presents the core equations with derivation sketches: the Cayley dispersion relation, modified dispersion relation for free particles, generalised uncertainty principle, and group-law defect. Section 4 gives physical interpretations. Section 5 performs consistency checks in the classical, continuum, and non-relativistic limits. Section **??** relates the proposal to loop quantum gravity [**?**, **?**], causal sets [**?**], and holographic approaches [**?**, **?**, **?**]. Sections **??**–**??** state bold conjectures, open problems, and conclusions.

2 Mathematical Framework

2.1 Hilbert-Space Setting

Let $(\mathcal{H}, \langle\cdot|\cdot\rangle)$ be a separable complex Hilbert space and let H be a densely defined self-adjoint operator on $\mathcal{H}$ (the Hamiltonian), with domain $\mathcal{D}(H) \subset \mathcal{H}$. We seek a family of operators $\{U_a\}_{a>0}$ approximating $e^{-iaH/\hbar}$ such that each U_a is a *unitary* operator on $\mathcal{H}$.

Definition 1 (Single-step rational evolution)**.** *A* rational single-step evolution scheme *is a map $U_a : \mathcal{H} \to \mathcal{H}$ of the form $U_a = f(aH/\hbar)$, where $f : \mathbb{R} \to \mathbb{C}$ is a rational function. It is called* exactly unitary *if $U_a^\dagger U_a = U_a U_a^\dagger = \mathbf{1}$ as an operator identity on $\mathcal{H}$, for every self-adjoint H and every $a > 0$.*

Definition 2 (Diagonal Padé approximant)**.** *For $n \geq 1$, the $[n, n]$ diagonal Padé approximant to e^{-iz} at $z = 0$ is*

$$e^{-iz} \approx R_n(z) \equiv \frac{P_n(-iz/2)}{P_n(+iz/2)}, \tag{2}$$

where $P_n(w) = \sum_{k=0}^{n} \binom{n}{k} \frac{(2n-k)!}{(2n)!} (2w)^k$ is the degree-n Padé numerator polynomial for e^{-w}. For $n = 1$ this gives $R_1(z) = (1 - iz/2)/(1 + iz/2)$, the Cayley map.

Definition 3 (Minimal temporal spectral triple)**.** *The* minimal temporal spectral triple *is the triple $(\mathcal{A}_T, \mathcal{H}_T, D_T)$ with:*

- *$\mathcal{A}_T = \mathbb{C}^2$ (commutative algebra, acts diagonally),*
- *$\mathcal{H}_T = \mathbb{C}^2$ (two-dimensional Hilbert space),*
- *$D_T = (\hbar/t_P)\,\sigma_x$, where $\sigma_x = \left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$.*

*This is the noncommutative analogue of a two-point space, the simplest nontrivial finite spectral triple for the time direction; compare the two-sheeted space appearing in the Connes–Chamseddine–Marcolli programme [**?**].*

2.2 Unitarity Classification Theorem

Theorem 1 (Unitarity classification)**.** *A rational function $f : \mathbb{R} \to \mathbb{C}$ yields a unitary operator $U_a = f(aH/\hbar)$ for* all *self-adjoint H on* all *Hilbert spaces if and only if $|f(x)| = 1$ for almost all $x \in \mathbb{R}$. Among rational functions, the unique family satisfying this condition and approximating e^{-ix} to order x^{2n+1} is the family of diagonal Padé approximants $R_n(x)$ of Definition 2. The $n = 1$ member is the* Cayley map*:*

$$f_{\text{Cay}}(x) = \frac{1 - ix/2}{1 + ix/2}, \qquad |f_{\text{Cay}}(x)| = 1 \quad \forall\, x \in \mathbb{R}. \tag{3}$$

Proof sketch. If H has an eigenvalue E, then U_a acts on the corresponding eigenstate as multiplication by $f(aE/\hbar) \in \mathbb{C}$. Unitarity requires $|f(aE/\hbar)| = 1$ for all $E \in \mathrm{Spec}(H)$. Since H is arbitrary, f must satisfy $|f(x)| = 1$ for all $x \in \mathbb{R}$. A rational function with $|f(x)| = 1$ on $\mathbb{R}$ must be a Blaschke product (or its reciprocal), which on the imaginary axis reduces to a ratio of conjugate polynomials: $f(x) = \overline{P(ix)}/P(ix)$. The approximation condition $f(x) = e^{-ix} + \mathcal{O}(x^{2n+1})$ uniquely selects the diagonal Padé family. The $n = 1$ case $f(x) = (1 - ix/2)/(1 + ix/2)$ is immediate. □

Comparison of schemes. Table 1 summarises the key properties of competing discretization schemes. The forward-Euler propagator $U_a = \mathbf{1} - iaH/\hbar$ has spectrum contained in a disk of radius greater than one for $E > 0$; norms grow in backward time. The backward-Euler propagator $U_a = (\mathbf{1} + iaH/\hbar)^{-1}$ has the opposite defect. Neither is unitary. The exact exponential $e^{-iaH/\hbar}$ is unitary but generically nonlocal. The Cayley scheme is the unique unitary, spatially local, rational approximant at minimal ($n = 1$) order.

Table 1: Comparison of discrete-time evolution schemes. SR compatibility is assessed at the level of physical predictions (Level 3 covariance; see Section 2.3).

Scheme	$f(x)$	Unitary	Group law	Spatially local	SR compat.
Forward Euler	$1 - ix$	No	No	Yes	No
Backward Euler	$(1 + ix)^{-1}$	No	No	Yes	No
Cayley $[1, 1]$	$\frac{1-ix/2}{1+ix/2}$	**Yes**	a^3 defect	Yes	Yes
Exact exponential	e^{-ix}	Yes	Yes	No	Yes
κ-Cayley	deformed	Yes	No	Yes	No (deformed)

2.3 Proper-Time Parameterisation and Lorentz Covariance

The central tension in any discrete-time scheme is that a fixed coordinate time step a is not Lorentz invariant: a boost with factor γ transforms it to $a' = \gamma a$. We resolve this by adopting the following:

Definition 4 (Level-3 Lorentz covariance)**.** *A discrete-time theory achieves* Level-3 Lorentz covariance *if all physical observables (probabilities, cross sections, S-matrix elements) computed from the theory are Lorentz invariant, even though the intermediate computational structure (the time lattice) is not manifestly covariant.*

Proper-time resolution. We interpret $a = t_P$ as a quantum of *proper time* along a particle's worldline, not coordinate time. The Planck time is a Lorentz scalar. Different inertial observers assign different coordinate durations to each proper-time tick:

$$\Delta t_{\text{coord}} = \gamma\, t_P = \frac{E}{m}\, t_P, \tag{4}$$

which is ordinary Lorentz time dilation. All observers agree on the total number of proper-time steps and on all invariant phase accumulations. This achieves Level-3 Lorentz covariance.

Massless limit. For photons, proper time does not advance along null geodesics. The correct treatment employs an affine parameter λ with $H_\lambda = |\hat{\boldsymbol{p}}|c$:

$$|\psi_\gamma(\lambda + \ell_P)\rangle = \frac{\mathbf{1} - \frac{i\ell_P}{2\hbar c}|\hat{\boldsymbol{p}}|c}{\mathbf{1} + \frac{i\ell_P}{2\hbar c}|\hat{\boldsymbol{p}}|c}\, |\psi_\gamma(\lambda)\rangle, \tag{5}$$

where $\ell_P = ct_P$ is the Planck length.

2.4 Derivation from the Noncommutative Spectral Triple

The deepest structural justification for the Cayley scheme is its derivation from the minimal temporal spectral triple (Definition 3).

Theorem 2 (Spectral triple selects Cayley scheme)**.** *Consider the product spectral triple* $D_{\text{full}} = D_{\text{space}} \otimes \mathbf{1}_2 + \gamma^5 \otimes D_T$ *with* $\{D_{\text{space}}, \gamma^5\} = 0$*. The Cayley transform of* D_{full}*,*

$$U = \frac{i\mathbf{1} + (t_P/\hbar)\, D_{\text{full}}}{i\mathbf{1} - (t_P/\hbar)\, D_{\text{full}}}, \tag{6}$$

restricted to forward-time (+) eigenstates of D_T *on an energy eigenstate* $|E\rangle$ *of* D_{space}*, yields*

$$U\big|_{E,+} = e^{-2i\arctan(t_P E/\hbar)} = e^{-i\omega_{\text{Cay}}(E)\cdot t_P}, \tag{7}$$

where $\omega_{\text{Cay}}(E) = (2/t_P)\arctan(t_P E/2\hbar)$ *is exactly the Cayley dispersion relation* (11)*.*

Proof sketch. On the eigenstate $|E,+\rangle$, D_{full} acts as multiplication by $E+\hbar/t_P$ (eigenvalue of D_T is $+\hbar/t_P$). The Cayley transform (6) then gives $U|_{E,+} = (i + t_P(E+\hbar/t_P)/\hbar)/(i - t_P(E+\hbar/t_P)/\hbar)$. A straightforward algebraic manipulation—noting that the $\hbar/t_P$ contribution produces an overall phase that cancels in the relative evolution—yields (7). □

Remark 1 (Spectral triple hierarchy)**.** *The Padé hierarchy corresponds to richer finite spectral triples:*

Finite spectral triple	*Points*	*Cayley transform gives*	*Physical scheme*
Minimal (S^0)	*2*	$[1,1]$ *Padé*	*Cayley/CN*
Next order	*3*	$[2,2]$ *Padé*	*Higher-order CN*
Continuum (S^1)	∞	*Exact exponential*	*Standard QM*

The physical selection principle—minimal nontrivial geometry—is the same principle that selects the Standard Model in the Connes–Chamseddine–Marcolli programme.

3 Core Equations and Results

3.1 The Fundamental Cayley Evolution Equation

The fundamental equation of the theory is (1), reproduced here for clarity with explicit $\hbar$:

$$\boxed{\left(\mathbf{1} + \frac{it_P H_\tau}{2\hbar}\right)|\psi(\tau + t_P)\rangle = \left(\mathbf{1} - \frac{it_P H_\tau}{2\hbar}\right)|\psi(\tau)\rangle.} \tag{8}$$

This is an implicit, linear equation relating the state at proper time $\tau + t_P$ to the state at τ. The evolution operator is

$$U^{\text{Cay}}_{t_P} = \left(\mathbf{1} + \frac{it_P H_\tau}{2\hbar}\right)^{-1}\left(\mathbf{1} - \frac{it_P H_\tau}{2\hbar}\right). \tag{9}$$

One verifies $U^\dagger U = \mathbf{1}$ immediately: $(U^{\text{Cay}})^\dagger U^{\text{Cay}} = \left[(\mathbf{1} - \frac{it_P H}{2\hbar})(\mathbf{1} + \frac{it_P H}{2\hbar})^{-1}\right]\left[(\mathbf{1} + \frac{it_P H}{2\hbar})^{-1}(\mathbf{1} - \frac{it_P H}{2\hbar})\right] = \mathbf{1}$, using $H^\dagger = H$ and the fact that $(\mathbf{1} \pm \frac{it_P H}{2\hbar})$ commute.

3.2 The Cayley Dispersion Relation

On an energy eigenstate $H|E\rangle = E|E\rangle$, the operator $U_{t_P}^{\text{Cay}}$ acts as multiplication by

$$e^{-i\omega_{\text{Cay}}(E)t_P} = \frac{1 - it_P E/2\hbar}{1 + it_P E/2\hbar}. \tag{10}$$

Solving for ω_{Cay} (taking the principal branch of arctan):

$$\boxed{\omega_{\text{Cay}}(E) = \frac{2}{t_P}\arctan\left(\frac{t_P E}{2\hbar}\right) = \frac{E}{\hbar} - \frac{t_P^2 E^3}{12\hbar^3} + \frac{t_P^4 E^5}{80\hbar^5} - \cdots} \tag{11}$$

The Taylor expansion in $t_P E/\hbar$ is obtained by using $\arctan z = z - z^3/3 + z^5/5 - \cdots$:

$$\begin{aligned}\omega_{\text{Cay}}(E) &= \frac{2}{t_P}\left[\frac{t_P E}{2\hbar} - \frac{1}{3}\left(\frac{t_P E}{2\hbar}\right)^3 + \frac{1}{5}\left(\frac{t_P E}{2\hbar}\right)^5 - \cdots\right] \\ &= \frac{E}{\hbar} - \frac{t_P^2 E^3}{12\hbar^3} + \frac{t_P^4 E^5}{80\hbar^5} - \cdots \end{aligned} \tag{12}$$

confirming the expansion in (11). The leading correction is $-t_P^2 E^3/(12\hbar^3)$ with *negative* sign.

Universality of sign. For any $[n, n]$ diagonal Padé scheme, the coefficient $c_2^{(n)}$ of $-t_P^2 E^3/\hbar^3$ satisfies $c_2^{(n)} > 0$ for all finite n, and $c_2^{(n)} \to 0$ as $n \to \infty$ (recovering the exact exponential). The sign of the leading correction is therefore *universal* across the entire unitary rational family:

Padé order $[n, n]$	$c_2^{(n)}$	Scheme
1 (Cayley)	1/12	Crank–Nicolson
2	1/180	Higher-order CN
$n \to \infty$	0	Exact $e^{-iHt/\hbar}$

3.3 Modified Dispersion Relation for Free Particles

For a massless particle ($E = pc$ in SI) evolving via (5), the Cayley dispersion gives the modified dispersion relation (MDR):

$$\boxed{E_\gamma^2 = p^2c^2\left(1 - \frac{p^2c^2}{6E_P^2}\right) + \mathcal{O}(E_P^{-4}),} \tag{13}$$

$$\boxed{v_g^{\text{Cay}} = \frac{\partial E_\gamma}{\partial p} = c\left(1 - \frac{E_\gamma^2}{4E_P^2}\right) + \mathcal{O}(E_P^{-4}),} \tag{14}$$

where $E_P = \hbar/t_P \approx 1.22 \times 10^{19}$ GeV is the Planck energy. The group velocity is *subluminal*: $v_g < c$.

For a massive particle:

$$E^2 = p^2c^2 + m^2c^4 - \frac{t_P^2}{6\hbar^2}\left(E^4 - m^4c^8\right) + \mathcal{O}(t_P^4), \tag{15}$$

which reduces correctly to $E^2 = p^2c^2 + m^2c^4$ for $E \ll E_P$.

Contrast with the κ-Cayley scheme. The κ-deformed Cayley dispersion at $\kappa = E_P$, computed via $\omega_{\kappa\mathrm{CN}} = (2/t_P)\arctan\left[\frac{1}{2}\sinh(t_P E/\hbar)\right]$, gives at leading order:

$$\omega_{\kappa\mathrm{CN}}(E) = \frac{E}{\hbar} + \frac{t_P^2 E^3}{12\hbar^3} + \mathcal{O}(t_P^4), \tag{16}$$

with a *positive* sign, implying *superluminal* group velocity. The derivation uses $\sinh\epsilon \approx \epsilon + \epsilon^3/6$ and $\arctan(\epsilon/2 + \epsilon^3/12) = \epsilon/2 + \epsilon^3/24 + \mathcal{O}(\epsilon^5)$, giving $\omega = E/\hbar + t_P^2 E^3/(12\hbar^3)$. The sign reversal between (11) and (16) is the key observable discriminator:

$$\Delta v/c = \begin{cases} -E^2/(4E_P^2) & \text{(Cayley, subluminal)}, \\ +E^2/(4E_P^2) & (\kappa\text{-Cayley, superluminal}). \end{cases} \tag{17}$$

3.4 Generalised Uncertainty Principle

The group velocity in the Cayley scheme is

$$v_g(E) = \frac{\partial\omega_{\mathrm{Cay}}}{\partial E}\cdot\hbar = \left(1 + \frac{t_P^2 E^2}{4\hbar^2}\right)^{-1}, \tag{18}$$

(in units where $c = 1$). This means the time for a wave packet to traverse a spatial interval Δx is $\Delta t = \Delta x/v_g = \Delta x(1 + t_P^2 E^2/4\hbar^2)$. Using the time-bandwidth uncertainty $\Delta\omega\cdot\Delta t \geq 1/2$ and $\Delta\omega = \Delta E/\hbar$, one obtains:

$$\boxed{\Delta E\cdot\Delta t \geq \frac{\hbar}{2}\left(1 + \frac{t_P^2 E^2}{4\hbar^2}\right) = \frac{\hbar}{2}\left(1 + \frac{E^2}{4E_P^2}\right).} \tag{19}$$

Limit checks:

- $E \ll E_P$: $\Delta E\cdot\Delta t \geq \hbar/2$ (standard Heisenberg).
- $E \sim E_P$: right-hand side $\geq 5\hbar/8$ (25% enhancement).
- $E \to \infty$: right-hand side $\to \infty$ (sub-Planck time resolution impossible).

3.5 The Group-Law Defect

The Cayley scheme does not satisfy the composition law $U_{2t_P}^{\mathrm{Cay}} = (U_{t_P}^{\mathrm{Cay}})^2$. On an energy eigenstate, the accumulated phase after two steps is $2\arctan(t_P E/2\hbar)$ (from two applications of $U_{t_P}^{\mathrm{Cay}}$), whereas $U_{2t_P}^{\mathrm{Cay}}$ contributes phase $2\arctan(t_P E/\hbar)$. The defect is:

$$\delta\theta(E) \equiv \theta_{\mathrm{Cay}}(2t_P, E) - 2\theta_{\mathrm{Cay}}(t_P, E) = 2\arctan\left(\frac{t_P E}{\hbar}\right) - 4\arctan\left(\frac{t_P E}{2\hbar}\right) = \frac{1}{2}\left(\frac{t_P E}{\hbar}\right)^3 + \mathcal{O}\left(\frac{t_P^5 E^5}{\hbar^5}\right), \tag{20}$$

obtained by expanding the arctangents to third order. The defect is *third order* in $t_P E/\hbar$, hence negligible for $E \ll E_P$. One may interpret it as an effective temporal curvature:

$$\Omega_{\mathrm{time}}(E) \equiv \frac{i\hbar}{t_P E}\ln\left(\frac{U_{2t_P}}{(U_{t_P})^2}\right) \sim \frac{t_P^2 E^2}{2\hbar^2}, \tag{21}$$

suggestive of an intrinsic curvature of the discrete temporal vacuum at the Planck scale.

3.6 Covariant Scalar Field Action

For completeness we state the Lorentz-covariant discrete action for a free scalar field on the Planck-scale hypercubic lattice with spacing ℓ_P in all four directions (metric signature $(+---)$):

$$\boxed{S_{\text{latt}} = \ell_P^4 \sum_{x \in \ell_P \mathbb{Z}^4} \left[\frac{1}{2} \sum_{\mu=0}^{3} \left(\frac{\phi(x + \ell_P \hat{\mu}) - \phi(x - \ell_P \hat{\mu})}{2\ell_P} \right)^2 - \frac{m^2 c^2}{2\hbar^2} \phi(x)^2 \right].} \tag{22}$$

The momentum-space propagator is

$$G(k) = \left[\sum_{\mu=0}^{3} \frac{\sin^2(k_\mu \ell_P)}{\ell_P^2} + \frac{m^2 c^2}{\hbar^2} \right]^{-1}, \tag{23}$$

and the on-shell dispersion $G^{-1}(k) = 0$ expanded for small $k_\mu \ell_P$ yields

$$E^2 = p^2 c^2 \left(1 - \frac{\ell_P^2 p^2}{3\hbar^2} \right) + \mathcal{O}(\ell_P^4), \tag{24}$$

consistent with (13) up to the numerical coefficient (1/3 vs. 1/6), the difference arising because the lattice action includes spatial lattice corrections in addition to the temporal Cayley ones.

4 Physical Interpretation

4.1 The Cayley Evolution as a Clock

Equation (8) may be read as a Planck-scale digital clock: at each tick of duration t_P in proper time, the quantum state advances by the unitary map $U_{t_P}^{\text{Cay}}$. The implicit nature of the equation—the state at $\tau + t_P$ appears on the left-hand side inside H_τ—is precisely the source of exact unitarity: it prevents the numerical amplification or dissipation associated with explicit (forward- or backward-Euler) schemes.

4.2 Proper Time as the Physical Clock Variable

The use of proper time τ rather than coordinate time t as the discretisation variable is not merely a convention; it is the physical implementation of the equivalence principle at the quantum level. Each massive particle carries its own proper-time clock that ticks at the universal rate t_P^{-1} in its instantaneous rest frame. The coordinate-time spacing $\Delta t = \gamma t_P$ that different observers assign to this tick is simply kinematic time dilation: there is no physical content in the frame-to-frame variation beyond ordinary special relativity [**?**].

4.3 Subluminal Propagation and Physical Causality

The group-velocity correction (14) is subluminal: $v_g < c$. This is physically benign—subluminal propagation preserves causality—and distinguishable from κ-deformed models, which predict superluminal propagation (17). The physical interpretation is that the discrete proper-time lattice introduces a small amount of "temporal drag": very-high-energy wave packets accumulate phase slightly more slowly than the continuum prediction, causing them to propagate slightly more slowly.

4.4 The GUP and Minimal Measurable Time

The GUP (19) encodes the impossibility of measuring time intervals shorter than t_P in the Cayley framework. At $E \sim E_P$, the uncertainty bound is enhanced by 25%; as $E \to \infty$, it diverges, reflecting that arbitrarily fine temporal resolution would require arbitrarily high energy, which in a theory with Planck-scale discreteness is physically impossible. This is analogous to Heisenberg's original gedanken argument for the position–momentum uncertainty principle, now applied to the time–energy pair with a Planck-scale regulator.

4.5 The Group-Law Defect and Temporal Curvature

The group-law defect (20) signals that the discrete proper-time lattice is not a group: the algebra of time translations is deformed at order $t_P^2 E^2/\hbar^2$. The effective temporal curvature (21) suggests a connection to quantum gravity: the Planck-scale vacuum may be characterised by an intrinsic temporal curvature of order unity (in Planck units), which is invisible at accessible energies but may leave imprints in Planck-scale scattering or in the very early universe [**?**].

4.6 Operational Meaning of the Covariant Lattice Action

The lattice action (22) is the standard Euclidean-rotated lattice field theory [**?**] evaluated at lattice spacing ℓ_P. Its propagator (23) is the natural generalisation of the continuum Feynman propagator, and the MDR (24) emerges as a *kinematic* consequence of the lattice geometry, not a dynamical input. Adding interactions (e.g., $\lambda\phi^4$ vertices) maintains path-integral unitarity, providing a consistent interacting extension of the free-field Cayley scheme.

5 Consistency Checks

5.1 Continuum Limit

For $t_P E/\hbar \ll 1$, the dispersion (11) reduces to $\omega_{\text{Cay}} \to E/\hbar$, the group velocity (14) to $v_g \to c$, and the GUP (19) to $\Delta E \cdot \Delta t \geq \hbar/2$. Standard quantum mechanics and relativistic kinematics are recovered exactly. The arctan function is smooth and analytic for all finite arguments, so no pathology arises in passing to the continuum.

5.2 Non-Relativistic Limit

For a non-relativistic massive particle with kinetic energy $E_k = p^2/2m \ll mc^2$, the rest-frame Hamiltonian is $H_\tau \approx mc^2 + p^2/2m$. The Cayley dispersion (11) gives:

$$\omega_{\text{Cay}} \approx \frac{mc^2}{\hbar} + \frac{p^2}{2m\hbar} - \frac{t_P^2 (mc^2)^2 p^2}{12\hbar^3 m} + \cdots \tag{25}$$

The Planck correction is suppressed by $(mc^2/E_P)^2$. For an electron, $(m_e c^2/E_P)^2 \approx (5.1 \times 10^{-4}\ \text{GeV}/1.22 \times 10^{19}\ \text{GeV})^2 \approx 1.7 \times 10^{-44}$, entirely beyond any conceivable measurement.

5.3 Dimensional Analysis

All correction terms in (11)–(19) are dimensionally consistent. The expansion parameter is the dimensionless ratio $t_P E/\hbar = E/E_P$. In SI units, the first correction to the dispersion relation is $-t_P^2 E^3/(12\hbar^3) \equiv -(E/E_P)^2 \cdot E/(12\hbar)$, confirming that it vanishes in the limit $t_P \to 0$ (continuum), $G \to 0$, or $E \to 0$ (low energy), as required.

5.4 Symmetry Analysis

Time-reversal. Under $\tau \to -\tau$ (time reversal), $U_{t_P}^{\mathrm{Cay}} \to (U_{t_P}^{\mathrm{Cay}})^{-1} = (U_{t_P}^{\mathrm{Cay}})^\dagger$. Since the scheme is unitary, this is equivalent to the antiunitary time-reversal operator of standard quantum mechanics. Time-reversal symmetry is preserved.

CPT invariance. The Cayley evolution is generated by the same Hamiltonian as continuum QM; no additional CPT-violating operators are introduced. CPT invariance is maintained to all orders in t_P.

Parity. The dispersion relation (11) depends on E^2 at leading correction order (through E^3 after multiplying by $E/\hbar$), and the group velocity is a function of E^2, so no parity-odd correction appears.

5.5 Consistency with Existing Observational Bounds

The highest-energy particles directly observed are ultra-high-energy cosmic rays (UHECR) at $E \sim 3 \times 10^{20}$ eV $= 3 \times 10^{11}$ GeV. The ratio $E_{\mathrm{UHECR}}/E_P \approx 2.5 \times 10^{-8}$, giving quadratic MDR corrections of order $(E/E_P)^2 \approx 6 \times 10^{-16}$. Current Lorentz invariance tests constrain quadratic MDR at the level $|\xi_2| \lesssim 10^{-8}$ from GRB and UHECR observations [**?**]. Our prediction, $\xi_2 = -1/4$, lies ~ 8 orders of magnitude below the sensitivity floor and is therefore consistent with all existing bounds.

The GRB time delay for a photon of energy $E_\gamma = 10$ GeV propagating over $L = 10^{26}$ m is

$$\Delta t_{\mathrm{GRB}} = \frac{L}{c} \cdot \frac{E_\gamma^2}{4E_P^2} \approx \frac{10^{26}}{3 \times 10^8} \cdot \frac{(10^{10} \times 1.6 \times 10^{-19})^2}{4 \times (1.22 \times 10^{19} \times 1.6 \times 10^{-10})^2} \approx 5 \times 10^{-20} \text{ s.} \tag{26}$$